ANALYSIS I — SHEET 6 (for Week 8 tutorials)
Tests for Series Convergence

1. In lectures it was shown that

$$\left(1+\frac{1}{n}\right)^n \to e \quad \text{as} \quad n \to \infty.$$

Deduce that

$$\left(1-\frac{1}{n}\right)^n \to \frac{1}{e} \quad \text{as} \quad n \to \infty.$$

2. For which of the following a_n is $\sum a_n$ convergent? Justify your answers.

$$\text{(i)} \quad \frac{n^4+\sin n}{2^n}; \qquad \text{(ii)} \quad \frac{n+2^n}{n2^n}; \qquad \text{(iii)} \quad \frac{n!}{n^n};$$

$$\text{(iv)} \quad (-1)^{n-1}\frac{\log n}{\sqrt{n}}; \qquad \text{(v)} \quad \frac{1}{n}\sin\left(\frac{1}{n}\right); \qquad \text{(vi)} \quad \left(1-\frac{1}{n}\right)^{n^2}.$$

3. Give either a proof, or a counter-example, to each of the following statements.

(i) If $n^2 a_n \to 0$ as $n \to \infty$ then $\sum a_n$ converges.
(ii) If $na_n \to 0$ as $n \to \infty$ then $\sum a_n$ converges.
(iii) If $\sum a_n$ converges, then $\sum {a_n}^2$ converges.
(iv) If $\sum a_n$ converges absolutely, then $\sum {a_n}^2$ converges.
(v) If $\sum a_n$ converges absolutely, then $|a_n| < 1/n$ for all sufficiently large n.
(vi) If $\sum a_n$ converges and $a_n/b_n \to 1$, then $\sum b_n$ converges.

4. Calculate exactly the following infinite sums. [*Hint: for the second sum, recall* $\cos n = \operatorname{Re} e^{in}$.]

$$\text{(i)} \quad \sum_{n=2}^{\infty}\frac{1}{n^2-1}; \qquad \text{(ii)} \quad \sum_{n=0}^{\infty}\frac{\cos n}{2^n}; \qquad \text{(iii)} \quad \sum_{n=0}^{\infty}\frac{n^2+7n+11}{n!}.$$

5. For what $\alpha, \beta > 0$ do the following series converge? You will need to consider the series for different cases of α and β.

$$\text{(i)} \quad \sum_{n=1}^{\infty}\frac{1}{n^\alpha(1+\alpha^{-n})}, \qquad \text{(ii)} \quad \sum_{n=2}^{\infty}\frac{1}{n^\alpha(\log n)^\beta}.$$

6. Let $\gamma_n = (\sum_{k=1}^{n} 1/k) - \log n$. By considering $\int_n^{n+1} 1/x \, \mathrm{d}x$, or otherwise, show that $0 < \gamma_{n+1} < \gamma_n$, and deduce that (γ_n) converges.

Let $t_n = \sum_{k=1}^{n} (-1)^{k-1}/k$. Show that $t_{2n} = \gamma_{2n} - \gamma_n + \log 2$, and deduce that

$$\sum_{r=1}^{\infty}(-1)^{k-1}/k = \log 2.$$

By considering the partial sums to $4n$ terms, or otherwise, evaluate the infinite sum

$$1+\frac{1}{3}+\frac{1}{5}-\frac{1}{2}+\frac{1}{7}+\frac{1}{9}+\frac{1}{11}-\frac{1}{4}+\frac{1}{13}+\frac{1}{15}+\frac{1}{17}-\frac{1}{6}+\cdots.$$

Extension Exercises (Optional)

7. (*Cauchy Product of Series)* (i) Let $\sum_0^\infty a_k$ and $\sum_0^\infty b_k$ be two absolutely convergent series and define

$$c_n = \sum_{k=0}^{n} a_k b_{n-k}.$$

Show that $\sum_0^\infty c_n$ is absolutely convergent and that

$$\sum_{n=0}^{\infty} c_n = \left(\sum_{k=0}^{\infty} a_k\right)\left(\sum_{k=0}^{\infty} b_k\right).$$

(ii) Give an example to show that $\sum_0^\infty c_n$ need not converge if $\sum_0^\infty a_k$ and $\sum_0^\infty b_k$ are only (conditionally) convergent.

(iii) Show for $|x| < 1$ that

$$(1-x)^{-2} = \sum_{n=0}^{\infty} (n+1)\, x^n.$$

8. (i) (*Cauchy Root Test*) Let a_n be a complex sequence and set $r = \limsup \sqrt[n]{|a_n|}$. Show that $\sum a_n$ converges if $r < 1$ and $\sum a_n$ diverges if $r > 1$. Give examples to show the test is inconclusive if $r = 1$.

(ii) (*Dirichlet Test*) Let $\sum a_n$ be a complex series with bounded partial sums and let (b_n) be a decreasing sequence with tends to 0. Show that $\sum a_n b_n$ converges.

9. (*Infinite Products*) Let a_k be a real sequence of non-zero terms. Let $\pi_n = a_1 a_2 \cdots a_n$. If $\pi_n \to L$ we write $L = \prod_1^\infty a_k$.

(i) Say that $a_n > 0$ for all n. Show that $\prod_1^\infty (1 + a_n)$ converges if and only if $\sum_1^\infty a_n$ converges.

(ii) Evaluate $\prod_2^\infty \left(1 - \frac{1}{n^2}\right)$.

(iii) Show that $\prod_2^\infty \left(1 - \frac{1}{n^3}\right)$ converges to a positive limit.

10. (*Cesàro Summability*) Let s_n denote the nth partial sum of the series $\sum_1^\infty a_k$ and set

$$\sigma_n = \frac{s_1 + s_2 + \cdots + s_n}{n}.$$

The series $\sum_1^\infty a_k$ is said to be Cesàro summable, with Cesàro sum S, if $\lim \sigma_n = S$.

(i) Show that $1 - 1 + 1 - 1 + \cdots$ has Cesàro sum $\frac{1}{2}$.

(ii) Show that $1 + 2 + 3 + \cdots$ is not Cesàro summable.

(iii) Show that if $\sum_1^\infty a_k = S$, then $\sum_1^\infty a_k$ has Cesàro sum S.

11. (*Wallis Product*) (i) By finding a suitable reduction formula, show that

$$\int_0^{\pi/2} \cos^{2n} x \, \mathrm{d}x = \frac{(2n)!\pi}{(n!)^2\, 2^{2n+1}}, \qquad \int_0^{\pi/2} \cos^{2n+1} x \, \mathrm{d}x = \frac{2^{2n}\,(n!)^2}{(2n+1)!}.$$

(ii) Deduce that

$$\binom{2n}{n}^2 \frac{\pi n}{2^{4n}} \to 1 \quad \text{as} \quad n \to \infty.$$

(ii) Hence prove Wallis' product

$$\frac{2}{\pi} = \frac{1}{2} \times \frac{3}{2} \times \frac{3}{4} \times \frac{5}{4} \times \frac{5}{6} \times \frac{7}{6} \times \frac{7}{8} \times \cdots.$$